MIL-C-85038
27 May 1976
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

CLOTH, KNITTED, ARAMID,
HIGH TEMPERATURE RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers the requirements for waffle type knit and for 1 x 1 rib knit aramid fabrics.

1.2 Classification - The knitted cloth covered by this specification shall be of the following types (see 6.2):

Type I - Waffle
Type II - 1 x 1 Rib

2. APPLICABLE DOCUMENTS

2.1 Issues of documents - The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FED-STD-191	Textile Test Methods

FSC 8305

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1491	Glossary of Knitting Imperfections

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FEDERAL TRADE COMMISSION

Rules and Regulations under the Textile Fiber Products Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 First article - When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.4 and 6.3).

3.2 Material - The cloth shall be made of high temperature resistant aramid yarn which does not char at a temperature less than 675°F (357°C) when tested as specified in 4.6.

3.2.1 Type I - The type I cloth shall be a waffle type, circular thermal knit conforming to the following construction:

1st Course Knitted	XXOO
2nd Course Knitted	..OO
3rd Course Knitted	XXOO
4th Course Knitted	XX..
5th Course Knitted	XX..
6th Course Knitted	XX..
7th Course Knitted	XXOO
8th Course Knitted	XXOO

NOTE: Knit: X Purl: O Tuck:

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3.2.2 Type II - The type II cloth shall be a circular knit having a 1 by 1 rib.

3.3 Chemical and physical properties - When tested as specified in 4.6, the cloth shall conform to the requirements specified in 3.2 and table I.

TABLE I. Chemical and physical properties

Characteristics	Requirements		Test Paragraph
	Type I	Type II	
Staple denier, nominal	1.5	1.5	4.6
Staple length, inch	1.5 - 2.0	1.5 - 2.0	4.6
Yarn count (cotton system), singles	18 ± 1.0	18 ± 1.0	4.6
Weight, oz. per square yard	7.5 ± 0.5	7.0 min.	4.6
Texture per inch, min.			
Wales	36	40	4.6
Courses	-	30	4.6
Repeats/6 inches, min.	28	-	4.6
Bursting strength, pounds, min.	80	90	4.6
Shrinkage, percent, max.			
Wales	12	12	4.6
Courses	6	6	4.6
pH	5.0 - 8.0	5.0 - 8.0	4.6
Thickness, mils, min.	30	30	4.6

3.4 Width - The width of the cloth shall be as specified by the procuring activity (see 6.2).

3.5 Color - Unless otherwise specified by the procuring activity, the color shall be natural.

3.6 Length and put up - Unless otherwise specified, the cloth shall be furnished in continuous lengths, each not less than 40 yards, and shall be put up in rolls as specified in PPP-P-1133.

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3.7 Fiber identification - Each roll shall be labeled, ticketed, and invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship - The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilty for inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance - Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections - The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4)

2. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable paragraphs of this specification.

4.4 First article inspection - When required (see 6.2), the first article sample submitted in accordance with 3.1 shall be visually inspected for appearance and color. The sample shall be tested in accordance with the methods specified in 4.6.

4.5 Quality conformance inspection - Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.5.1 Examination of the end item - Examination of the end item shall be in accordance with 4.5.1.1 through 4.5.1.4.

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4.5.1.1 Yard-by-yard examination - The required yardage of each roll shall be examined and visual defects as defined in MIL-STD-1491 shall be classified as listed in Table II. All defects found shall be counted regardless of their proximity one to another, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The sample unit for this examination shall be one linear yard. The sample size shall be in accordance with Inspection Level II of MIL-STD-105. The AQL shall be 2.5 major defects and 6.5 total defects per 100 units (yards). The lot size shall be expressed in units of one linear yard each. An approximate equal number of yards shall be examined from each roll selected in accordance with Table III.

TABLE II. Classificaton of defects

Defects	Classification	
	Major	Minor
Hole, cut, tear, any	X	
Run or dropped stitch	X	
Float, pulled stitch or needle streak		X
Mend	X	
Abrasion Mark 1/	X	
Thin place 1/	X	
Thin yarn 1/	X	
Tight yarn		X
Slubs or slug, more than twice the size of normal yarn		X
Foreign matter 1/		X
Knots extending above the surface of cloth 1/		X
Kinks or untrimmed ends		X
Crease or wrinkle (embedded)		X
Stop mark	X	
Shade bar 1/		X
Spot or stain	X	

1/ Clearly visible at normal inspection distance (3 feet).

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4.5.1.2 Overall examination - Each defect listed below shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size and the acceptance number shall be as specified in table III.

OVERALL DEFECTS

Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the piece.
Off shade, shaded end to end, side to side, side to center or throughout the roll.
Poor dye penetration, mottled, streaky or cloudy.
Overall width less than specified.

4.5.1.3 Examination for length

4.5.1.3.1 Individual rolls - During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The maximum number of defects acceptable in the sample shall be in accordance with table III.

TABLE III. Sample size

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,0001 up to and including 35,000	13	0
35,001 up to and including 150,000	20	1
150,001 and over	32	2

1/ If lot is fewer than three rolls, each roll in the lot shall be examined.

4.5.1.3.2 Total yardage in sample - The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for the examination of individual rolls.

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4.5.1.4 Examination for compliance with Textile Fiber Products Identification Act - During the yard-by-yard examination, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.5.2 Packaging inspection - The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of PPP-P-1133.

4.6 Methods of examination and test - The methods of testing specified in FED-STD-191, wherever applicable and as listed in table IV shall be followed. The physical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 3-½ yards (full width) of the finished cloth. The lot size shall be expressed in units of one yard each. The lot shall be unacceptable if one or more units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sampling size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. Methods of examination and test

Characteristics	Requirement paragraph	Fed-Std-191 Test method as applicable
Staple		
Denier	Table I	1/
Length	Table I	1/
Yarn count	Table I	1/
Char point	3.2	1/
Weight	Table I	5041
Material identification	3.2	1/
Texture (wales and courses)	Table I	5070 3/
Repeats	Table I	4/

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TABLE IV. Methods of examination and test (Con't)

Characteristic	Requirement paragraph	Fed-Std-191 Test method as applicable
Bursting strength	Table I	5120
Knit	3.2.1, 3.2.2	Visual
Shrinkage in laundering Wales Course	Table I	5556 2/
pH	Table I	2811
Thickness	Table I	5030

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Cotton laundering procedure shall be followed.

3/ When counting wales, count the total knit and purl stitches per inch of the cloth construction.

4/ Count the number of repeats (dial tuck) in six inches of cloth. The number of repeats shall be the average of five determinations (to the nearest whole number) per sample unit.

5. PACKAGING

5.1 Preservation-packaging - Preservation-packaging shall be Level A or C as specified (see 6.2).

5.1.1 Levels A and C - The cloth, put up as specified in 3.6, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing - Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C - The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking - In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1133.

6. NOTES

6.1 Intended use - The cloth is intended for use in the manufacture of thermal underwear.

6.2 Ordering data - Procurement documents should specify the following:

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(a) Title and date of this specification

(b) Type and color required (see 1.2 and 3.5)

(c) Width required (see 3.4)

(d) Length required (see 3.6)

(e) Levels of preservation-packaging and packing (see 5.1 and 5.2)

(f) When first article is required (see 3.1, 4.4 and 6.3)

(g) Quantity required

6.3 First article - When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample consisting of 3-1/2 yards (full width) of the finished cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Supersession data - This specification supersedes Naval Air Systems Command Purchase Description AS3613, dated 16 January 1973.

Custodians:
Navy - AS
Air Force - 82

User activity:
Navy - SA

Preparing Activity:
Navy - AS
(Project No. 8305-0517)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE CLOTH, KNITTED, ARAMID, HIGH TEMPERATURE RESISTANT
MIL-C-85038

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4 REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

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